electronics

MDPI

Article

Trust-Aware Federated Learning for Privacy-Preserving IoT Intrusion Detection

Santosh Kumar Banbhrani [1,2,*], Chen Wei [1] and Muhammad Asad [3,*]

[1] School of Electrical Engineering, Shaoyang University, Shaoyang 422000, China; 2569@hnsyu.edu.cn
[2] Department of Information and Computing, University of Sufism and Modern Science, Bhit Shah 70140, Sindh, Pakistan
[3] Faculty of Science and Engineering, University of Plymouth, Plymouth PL4 8AA, UK
* Correspondence: santosh.kumar@hnsyu.edu.cn (S.K.B.); muhammad.asad@plymouth.ac.uk (M.A.)

Abstract

The rapid growth of Internet of Things (IoT) networks has increased the need for intrusion detection mechanisms that provide accurate threat detection while preserving privacy, resisting malicious clients, and limiting communication overhead. Existing Federated Learning (FL)-based intrusion detection systems (IDS) remain challenged by heterogeneous non-IID client data, exposure of model updates, adversarial client participation, and frequent model transmission. To address these limitations, this paper proposes FLZT-IDS (Federated Learning with Zero-Trust for Intrusion Detection in IoT). The framework employs gradient-deviation monitoring to mitigate the effects of heterogeneous client updates, while Graph Neural Networks (GNN) and an Attention-based BiLSTM capture relational and temporal attack patterns, respectively. Secure Aggregation with Local Differential Privacy (SALDP) protects transmitted model updates, whereas dynamic trust-based client filtering limits the contribution of potentially malicious participants. An adaptive communication mechanism transmits local updates only when the model change exceeds a predefined threshold, thereby reducing unnecessary communication. Experiments on CICIDS2017 and Edge-IIoTset show that FLZT-IDS achieves 97.8% and 96.2% accuracy, respectively. Compared with FL-IDS, the adaptive communication and trust-filtering mechanisms reduce cumulative communication overhead by more than 30% after 100 communication rounds. Under model-poisoning experiments with 30% malicious clients, FLZT-IDS maintains accuracies of 89.3% on CICIDS2017 and 87.1% on Edge-IIoTset, demonstrating improved robustness under the evaluated adversarial setting.

Keywords: federated learning; Zero Trust security; intrusion detection; privacy-preserving IoT; threat detection

Check for updates

Academic Editor: Arkaitz Zubiaga

Received: 3 August 2026
Revised: 19 September 2026
Accepted: 25 September 2026
Published: 9 October 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

The rapid proliferation of the Internet of Things (IoT) has transformed modern digital ecosystems by enabling seamless connectivity across diverse domains such as smart cities, healthcare, industrial automation, and transportation [1]. With billions of IoT devices deployed globally, these interconnected systems generate vast amounts of sensitive data and require real-time communication. However, the expansion of IoT infrastructure has also introduced significant cybersecurity risks, making these networks prime targets for malicious actors. Threats such as Distributed Denial of Service (DDoS) attacks, Man-in-the-Middle (MITM) exploits, malware infections, and unauthorized access compromise not only data integrity but also the reliability of critical services [2].

Intrusion Detection Systems (IDS) have emerged as a key defense mechanism to mitigate cybersecurity risks in IoT networks. Traditional IDS models rely on centralized architectures, where data collected from edge devices is transmitted to a central server for analysis [3]. However, this centralized approach presents multiple challenges, including scalability issues, single points of failure, and heightened privacy risks. The massive volume of IoT-generated data places a significant burden on network bandwidth, while central data repositories become attractive targets for cyber adversaries.

Machine learning-based IDS solutions have gained traction due to their ability to identify complex attack patterns [4]. However, conventional machine learning approaches require access to extensive labeled datasets, often collected and stored in centralized repositories. This raises privacy concerns, as raw data must be transmitted to third-party servers for model training. Additionally, centralized models are vulnerable to adversarial attacks, where manipulated training data can deceive the system into misclassifying threats [5]. The lack of robust security frameworks within traditional IDS models exacerbates these challenges, necessitating a paradigm shift toward decentralized and privacy-preserving intrusion detection mechanisms.

Federated Learning (FL) has emerged as a promising approach to addressing these concerns by enabling collaborative model training across multiple devices without sharing raw data. In FL, local models are trained on edge devices, and only model updates (e.g., gradients or parameters) are shared with a central server, thereby preserving data privacy. Despite its advantages, FL-based IDS solutions face several challenges, including non-IID data distributions, high communication overhead, vulnerability to model poisoning attacks, and lack of trust assurance [6]. The inherent nature of FL assumes that all participating devices contribute honestly, yet adversarial participants can compromise global model integrity through poisoning attacks. Additionally, securing communication during federated updates remains a critical challenge.

From a methodological perspective, anomaly and fault detection problems have been addressed through different learning paradigms, including supervised deep learning, few-shot and cross-modal representation learning, and distributed federated learning. Recent approaches demonstrate the value of specialized learning paradigms for complex anomaly detection. For example, a study [7] proposed BCLIP-ADer, a few-shot vision–language anomaly detection framework that combines Bayesian prompt learning with cross-modal feature refinement to improve generalization to previously unseen anomalies. Similarly, in [8] the authors developed a meta-learning-based graph convolutional wavelet network that combines multi-scale signal decomposition, graph-based dependency modeling, and adaptive loss learning for complex dynamic systems. Although these approaches demonstrate the effectiveness of advanced representation learning in their respective domains, they are designed for centralized task-specific learning and do not address the combined requirements of privacy-preserving distributed training, malicious-client filtering, and communication-efficient model exchange that arise in federated IoT intrusion detection. These requirements motivate a learning framework that integrates complementary relational and temporal representations with privacy, trust, and communication-aware federated mechanisms.

To further enhance security, the Zero Trust Security Model has gained widespread adoption, advocating a "never trust, always verify" approach. Unlike traditional security models that rely on implicit trust within network perimeters, Zero Trust mandates continuous authentication and access verification for all entities, regardless of location or prior authentication [9]. Integrating Zero Trust principles with FL-based IDS can significantly improve the security posture of IoT environments by ensuring that only verified updates contribute to the global model while mitigating adversarial risks.

This paper proposes an innovative FL with Zero Trust for Secure and Private Intrusion Detection in IoT (FLZT-IDS) framework, integrating privacy-preserving mechanisms to address the aforementioned challenges. By leveraging Graph Neural Networks (GNN) for relational pattern recognition, Attention-based BiLSTM for sequential anomaly detection, and Secure Aggregation with Local Differential Privacy (SALDP), this approach ensures robust, scalable, and efficient intrusion detection in IoT networks.

1.1. Research Problem & Objectives

Despite the privacy-preserving benefits of FL, existing FL-based IDS struggle with key challenges: vulnerability to adversarial attacks, high communication overhead, and limited adaptability [10]. Traditional IDS models expose sensitive data by relying on centralized processing, while FL, though decentralized remains susceptible to model inversion and poisoning attacks [11]. Malicious clients can inject manipulated updates to compromise the global model, and frequent communication between edge devices and the server strains bandwidth in large-scale IoT networks [12]. Moreover, many FL-based IDS frameworks fail to adapt to evolving threats, resulting in high false positive and negative rates.

To overcome these limitations, this research introduces FLZT-IDS: a Zero-Trust-enabled, Privacy-Preserving FL-based IDS for IoT. The framework integrates GNN for relational pattern recognition and Attention-based BiLSTM for sequential anomaly detection [13]. A SALDP mechanism further enhances privacy and robustness. Experimental results on CICIDS2017 and Edge-IIoTset datasets confirm the superior accuracy, scalability, and resilience of FLZT-IDS over existing approaches. Table 1 summarizes the abbreviations and mathematical notations widely used throughout this manuscript.

Table 1. List of Abbreviations and Notations Used in This Paper.

Abbreviation/ Symbol	Definition	Abbreviation/ Symbol	Definition
FL	Federated Learning	IDS	Intrusion Detection System
IoT	Internet of Things	GNN	Graph Neural Network
BiLSTM	Bidirectional Long Short-Term Memory	DP	Differential Privacy
LDP	Local Differential Privacy	SALDP	Secure Aggregation with Local Differential Privacy
ZTM	Zero Trust Model	FPR	False Positive Rate
ACC	Accuracy	P	Precision
R	Recall	F1	F1-Score
FedAvg	Federated Averaging	FedGAT	Federated Graph Attention Network
Fed-ANIDS	Federated Anomaly-based Intrusion Detection System	FLZT-IDS	Federated Learning with Zero Trust for Intrusion Detection in IoT
SMOTE	Synthetic Minority Over-sampling Technique	HE	Homomorphic Encryption
CCPA	California Consumer Privacy Act	GDPR	General Data Protection Regulation
$\|D\|$	Total number of training samples	$\delta_{\max}$	Maximum allowable gradient-deviation threshold
e_t	Unnormalized temporal-attention score	I	Identity matrix in the Gaussian-noise covariance
y	Predicted binary class label (Benign or Attack)	$\oplus$	Homomorphic addition of encrypted updates
ε	Differential-privacy budget	$\varepsilon_{\text{comm}}$	Communication-update threshold

1.2. Contributions

This paper makes the following key contributions to IoT security research:

- We integrate GNN with FL to capture relational dependencies in network traffic, enhancing intrusion detection accuracy for complex cyber threats.
- We employ an Attention-based BiLSTM model to detect sequential attack patterns, improving adaptability to evolving threats and reducing false positive rates.
- We implement SALDP to ensure privacy-preserving model updates, mitigating risks from adversarial model poisoning attacks.
- We evaluate the proposed framework using CICIDS2017 and Edge-IIoTset datasets, demonstrating superior detection performance, privacy protection, and scalability compared to existing FL-based IDS solutions.

1.3. Paper Organization

The paper is structured as follows. Section 2 reviews literature review on IDS frameworks, FL in cybersecurity, and Zero Trust security. Section 3 presents the proposed FL-ZT-IDS methodology, integrating GNN, Attention-based BiLSTM, and SALDP. Section 4 outlines the experimental setup, including datasets, model training, and evaluation metrics. Section 5 analyzes results and compares FL-ZT-IDS with existing approaches. Section 6 discusses key findings, limitations, and future research directions. Finally, Section 7 concludes the paper.

2. Literature Review

This section reviews intrusion detection in IoT, covering traditional IDS approaches, FL-based IDS, and Zero Trust security. It also examines the limitations of centralized IDS and the need for privacy-preserving, adaptive security solutions.

2.1. Traditional IDS Approaches for IoT Security

IDS play a critical role in securing IoT networks by identifying unauthorized activities and cyber threats. Traditional IDS approaches are broadly classified into signature-based detection and anomaly-based detection [14]. Signature-based IDS rely on predefined attack signatures, making them effective against known threats but incapable of detecting novel cyberattacks [15]. In contrast, anomaly-based IDS monitor network behavior and flag deviations as potential threats, enabling the detection of previously unseen attacks but often resulting in higher false positive rates [16]. These conventional IDS models face limitations in adapting to evolving attack patterns, handling large-scale IoT networks, and maintaining real-time detection accuracy.

2.1.1. Signature-Based vs. Anomaly-Based Detection Methods

Signature-based IDS compare incoming network traffic against a database of known attack patterns. This approach is widely used in IDS tools such as Snort and Suricata, offering high detection accuracy for previously identified threats with minimal false positives [17]. However, the reliance on predefined signatures makes these systems ineffective against zero-day exploits and adversarial attacks [18]. To address this limitation, anomaly-based IDS employ machine learning, statistical models, and deep learning techniques to establish baselines of normal network behavior and detect deviations [19]. While anomaly-based IDS can identify unknown attacks, they suffer from high false positive rates and require frequent model retraining to adapt to dynamic IoT environments.

2.1.2. Challenges in Centralized Intrusion Detection Systems

Most traditional IDS frameworks operate in a centralized manner, where IoT devices transmit network logs to a central server for analysis. While this architecture facilitates large-scale monitoring, it introduces scalability, latency, and privacy challenges [20]. High network traffic from multiple IoT devices leads to computational bottlenecks, limiting real-

time detection capabilities. Moreover, single points of failure in centralized IDS make them vulnerable to distributed denial-of-service (DDoS) attacks and adversarial manipulations, reducing their effectiveness in securing IoT networks [21]. These limitations highlight the need for decentralized and privacy-preserving IDS solutions, such as FL-based IDS, which can enhance security without compromising efficiency.

2.2. Federated Learning for Intrusion Detection

FL has emerged as an effective approach for intrusion detection in IoT by enabling collaborative model training without exposing raw data, thus preserving privacy and reducing communication overhead. Unlike centralized IDS, where data is transmitted to a central server, FL ensures that only model updates are shared, reducing the risk of data breaches. This decentralized learning paradigm is particularly beneficial for IoT networks, where data is generated across heterogeneous devices with varying computational capabilities [20].

Several FL-based IDS frameworks have demonstrated improved detection accuracy while maintaining privacy. In [22] an FL-based IDS for IoT achieves efficient anomaly detection without data centralization. However, FL-based IDS faces challenges, including non-IID data distributions, which can degrade global model performance, and model poisoning attacks, where adversarial nodes inject malicious updates to compromise the system [23]. Additionally, FL introduces high communication costs, making real-time intrusion detection challenging in resource-constrained IoT environments. Addressing these issues requires optimized aggregation mechanisms, adversarial defense techniques, and communication-efficient training strategies to enhance the robustness and scalability of FL-based IDS [24].

2.3. Zero-Trust Security Model in IoT

The Zero-Trust Security Model (ZTSM) has gained significant attention in securing IoT environments by enforcing strict access controls and eliminating implicit trust in devices or users [25]. Unlike traditional perimeter-based security approaches, Zero-Trust requires continuous authentication, authorization, and policy enforcement to mitigate unauthorized access risks. Recent studies [26,27] have explored its integration into IoT networks, emphasizing its role in protecting against evolving cyber threats such as man-in-the-middle attacks, privilege escalation, and unauthorized device access. However, applying Zero-Trust principles to IoT remains challenging due to the heterogeneous nature of IoT devices, constrained computational resources, and the dynamic nature of network interactions [28].

Incorporating Zero-Trust with FL-IDS presents a promising avenue for enhancing security without compromising privacy. Studies [29,30] have demonstrated that Zero-Trust-driven FL-IDS improves intrusion detection accuracy by ensuring that only verified and authenticated devices participate in federated training. Additionally, secure aggregation techniques such as homomorphic encryption and differential privacy can further mitigate model poisoning risks while maintaining the decentralized advantage of FL. Despite these advancements, ensuring scalability and optimizing computational overhead remain open research challenges that require further investigation in large-scale IoT deployments.

2.4. Privacy-Preserving Techniques in FL-Based IDS

Privacy preservation is a critical concern in FL-IDS, as sensitive data remains distributed across IoT devices. Unlike centralized IDS, where raw data is transferred to a central server, FL ensures that only model updates are shared, reducing exposure to data breaches [31]. However, privacy threats such as model inversion attacks, membership inference attacks, and data reconstruction vulnerabilities still pose significant risks [32]. To

address these issues, various privacy-preserving techniques, including differential privacy (DP), homomorphic encryption (HE), and secure multi-party computation (SMPC), have been integrated into FL-based IDS frameworks [12]. DP adds statistical noise to model updates, ensuring that individual device contributions remain indistinguishable, while HE allows computations on encrypted data, preventing attackers from extracting sensitive information during aggregation [33].

Secure aggregation protocols further enhance privacy by masking individual model updates before they reach the central server, preventing malicious entities from reconstructing private data [34]. Recent research [35,36] has explored combining Zero-Trust principles with privacy-preserving FL-IDS to ensure that only authenticated nodes contribute to the training process, reducing risks from adversarial attacks. Despite these advancements, achieving an optimal balance between privacy, computational efficiency, and model accuracy remains an ongoing challenge, particularly in resource-constrained IoT environments where cryptographic operations introduce additional overhead. Future research must focus on lightweight encryption techniques and adaptive privacy mechanisms to enhance the deployment feasibility of privacy-preserving FL-based IDS at scale.

2.5. Learning Paradigms and Related Approaches

Existing anomaly and intrusion detection studies can also be distinguished according to their underlying learning paradigms. Supervised deep learning methods learn discriminative representations from labeled training data and can achieve high predictive performance when representative labels are available [37]. Recent graph-based approaches extend this paradigm by explicitly modeling structural dependencies. For example, Ref. [8] proposed a meta-learning-based graph convolutional wavelet network that combines multi-scale wavelet decomposition with graph convolution and adaptive loss learning to capture nonlinear relationships among multiple system variables. Although such approaches demonstrate the effectiveness of graph-based relational modeling, they assume centralized access to training data and are not designed to address privacy or malicious-client behavior in distributed learning environments.

Alternative paradigms have been developed for settings in which abnormal samples are scarce. Ref. [7] proposed BCLIP-ADer, a few-shot anomaly detection approach based on Bayesian prompt learning and contrastive vision–language representations. The method improves generalization to unseen anomaly types by learning diverse text prompts and refining cross-modal representations. Such few-shot and cross-modal approaches address limited anomaly examples and representation generalization; however, their objectives differ from federated intrusion detection, where model updates originate from distributed and potentially untrusted clients [38].

Federated learning provides a complementary paradigm by allowing multiple clients to collaboratively train a model without directly sharing raw data [39]. Existing FL-based IDS methods address different subsets of this problem. FL-IDS [40] employs federated training with a conventional neural-network-based detector, while FedGAT [41] introduces graph attention to capture relational dependencies among network traffic instances. Fed-ANIDS focuses on federated anomaly detection using autoencoder-based representations. These approaches improve decentralization or attack representation, but they do not jointly incorporate trust-aware client participation, local differential privacy with secure aggregation, and communication-triggered model transmission.

Therefore, the distinction of FLZT-IDS is not the individual use of GNN, BiLSTM, federated learning, or privacy mechanisms in isolation. Rather, the proposed framework integrates complementary relational and temporal modeling with privacy-preserving model-update transmission, dynamic trust-based client filtering, and adaptive communication

within a single federated IDS pipeline. Table 2 summarizes the methodological differences between representative approaches and clarifies the specific integration addressed by FLZT-IDS.

Table 2. Comparison of representative learning paradigms and their relationship to FLZT-IDS.

Method	Learning Paradigm	Primary Modeling Focus	Privacy-Aware FL	Trust/ Robustness	Adaptive Comm.
BCLIP-ADer	Few-shot vision–language learning	Cross-modal anomaly representation and unseen-anomaly generalization	No	No	No
Meta-learning graph convolutional wavelet network	Graph-based meta-learning	Multi-scale dynamic and relational modeling	No	No	No
FL-IDS	Federated learning	Distributed neural-network-based intrusion detection	Partial	No	No
FedGAT	Federated graph learning	Relational dependencies in network traffic	Partial	No	No
Fed-ANIDS	Federated anomaly detection	Autoencoder-based anomaly representation	Partial	No	No
FLZT-IDS	Federated hybrid learning	Relational and temporal intrusion modeling	Yes	Yes	Yes

2.6. Gaps in Existing Work & Research Direction

Despite significant advancements in FL, Zero-Trust Security, and IDS for IoT, several research gaps remain unaddressed. Existing FL-based IDS frameworks struggle with challenges such as high communication overhead, vulnerability to adversarial attacks, and handling non-IID data distributions [29,42]. Similarly, Zero-Trust Security models in IoT require extensive computational resources for continuous verification, making them less feasible for resource-constrained edge devices [43]. As summarized in Table 2, existing approaches address important but largely complementary aspects of anomaly and intrusion detection, including graph-based representation learning, few-shot anomaly generalization, federated training, and privacy preservation. Among the representative methods reviewed, none jointly combines relational and temporal intrusion modeling, privacy-preserving federated update transmission, dynamic trust-based client filtering, and adaptive communication. FLZT-IDS is designed to address this integration gap within a unified federated IoT intrusion-detection pipeline.

To address these challenges, future research should focus on lightweight privacy-preserving techniques, robust aggregation mechanisms for secure FL, and adaptive intrusion detection models that can evolve with emerging threats. Additionally, hybrid approaches integrating blockchain with FL and Zero-Trust could further enhance trust verification and data integrity in decentralized environments.

3. Proposed Methodology: FLZT-IDS

3.1. Overview of the Proposed Framework

The proposed FLZT-IDS integrates FL, Zero-Trust Security, and Privacy-Preserving mechanisms to enhance the security and efficiency of IoT-based intrusion detection. Unlike traditional IDS, which rely on centralized data collection, FLZT-IDS ensures decentralized training across IoT devices, reducing privacy risks while enabling real-time attack detection. As shown in Figure 1, the framework enforces strict access policies using Zero-Trust principles, ensuring that only authenticated and verified nodes participate in the FL process. Furthermore, privacy-preserving techniques, such as differential privacy and secure aggregation, safeguard model updates from adversarial threats and data inference attacks.

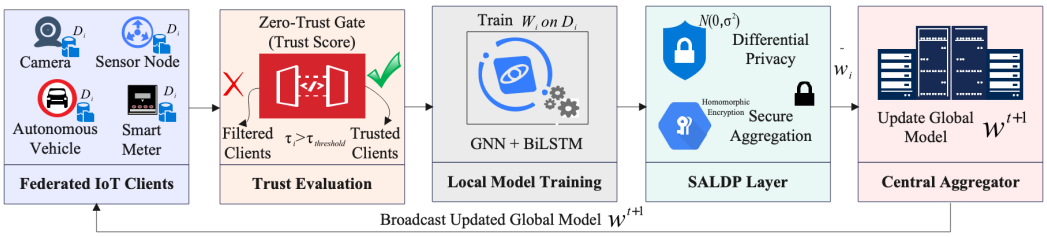


Figure 1. System overview of FLZT-IDS: IoT clients train local models on private data and are filtered by a Zero-Trust Gate based on trust scores. Trusted clients apply a multi-modal deep model (GNN + BiLSTM), followed by privacy-preserving transmission using SALDP. The central server aggregates these updates to form a global model W^{t+1}, which is broadcast back to clients.

Let $\mathcal{D} = \{D_1, D_2, \ldots, D_N\}$ represent the distributed datasets across N IoT devices. Each device i trains a local model w_i on its dataset D_i, and instead of sharing raw data, only model updates Δw_i are transmitted to the central server for aggregation. The global model update at round t follows:

$$w^{t+1} = \sum_{i=1}^{N} \frac{|D_i|}{|\mathcal{D}|} w_i^t \tag{1}$$

where $|D_i|$ is the number of training samples at device i, ensuring weighted aggregation proportional to dataset sizes.

To incorporate Zero-Trust Security, each participating IoT device must satisfy the server-side trust-verification criterion before contributing to the FL process. This is achieved by assigning each device a trust score τ_i, computed as:

$$\tau_i = \alpha R_i + \beta C_i + \gamma A_i \tag{2}$$

where R_i, C_i, and A_i are normalized to the range $[0, 1]$ and are computed by the server-side trust monitor rather than obtained directly from CICIDS2017 or Edge-IIoTset. In the experiments, the weighting coefficients are set to $\alpha = 0.4$, $\beta = 0.3$, and $\gamma = 0.3$. R_i represents the proportion of successful client participations within a sliding window of the previous $W = 5$ communication rounds, while C_i represents the proportion of protocol-compliance checks successfully satisfied during the same window. A_i is derived from the historical deviation of the client's model updates from the global training behavior, with smaller deviations producing higher confidence scores. All three components are computed from server-observed training and communication behavior and do not use the ground-truth attack labels of CICIDS2017 or Edge-IIoTset. Therefore, the trust calculation does not introduce label leakage.

The trust evidence is updated after each communication round using the most recent five-round window. For cold-start clients, an initial trust score of $\tau_i^0 = 0.7$ is assigned, after which the score is replaced by the observed evidence-based value. Since the trust evidence is computed by the server rather than self-reported by the client, a malicious client cannot directly forge R_i, C_i, or A_i; abnormal model-update behavior instead reduces its anomaly-confidence and compliance scores. A client is accepted for aggregation only when $\tau_i > \tau_{\text{threshold}}$.

A device is allowed to participate only if $\tau_i > \tau_{\text{threshold}}$, ensuring only trustworthy nodes contribute to the model.

For privacy preservation, we integrate differential privacy by adding noise $\mathcal{N}(0, \sigma^2)$ to each local update:

$$\tilde{w}_i = w_i + \mathcal{N}(0, \sigma^2) \tag{3}$$

where σ controls the privacy–accuracy tradeoff. Secure aggregation techniques ensure that the server cannot reconstruct individual updates, preserving confidentiality.

The combination of FL, Zero-Trust Security, and Privacy-Preserving Mechanisms ensures that FLZT-IDS provides robust, scalable, and attack-resilient intrusion detection in IoT environments. The subsequent sections elaborate on the detailed methodology for model training, security enforcement, and privacy-preserving techniques.

3.2. Federated Learning-Based IDS for IoT

The FLZT-IDS framework leverages FL to develop an IDS that enables decentralized model training while preserving privacy. Unlike centralized IDS, FL ensures that each IoT device trains a local model on its data and transmits only model updates to the central server. This reduces privacy risks and enhances scalability. However, FL-based IDS faces key challenges, including non-IID data distribution, susceptibility to adversarial attacks, and communication overhead, which impact model performance.

One of the critical challenges in FL is data heterogeneity, where IoT devices generate non-IID data, leading to biased global models [44]. To quantify this divergence, a device-specific gradient deviation factor is introduced:

$$\delta_i = ||\nabla L_i(w_i^t) - \nabla L(w^t)||^2 \tag{4}$$

where $\nabla L_i(w_i^t)$ is the local gradient at device i, and $\nabla L(w^t)$ is the aggregated gradient. If $\delta_i > \delta_{\max}$, additional local training is triggered to align the model with the global objective.

To counter adversarial updates, Zero-Trust Security mechanisms verify the integrity of model updates before aggregation. Each device's update is evaluated using its trust score τ_i (Equation (2)). Additionally, SALDP protects updates from reconstruction attacks by applying encryption and noise injection, as formulated in Equation (3).

To further optimize model communication efficiency, an adaptive update mechanism is introduced, where only significant updates are transmitted. The communication round is triggered when the model's local change Δw_i surpasses a predefined threshold:

$$||\Delta w_i|| > \epsilon \tag{5}$$

where ϵ controls the frequency of updates, reducing unnecessary transmissions.

By integrating gradient alignment, Zero-Trust verification, secure aggregation, and adaptive communication, FLZT-IDS ensures efficient, scalable, and attack-resilient intrusion detection for IoT networks.

3.3. GNN for Relational Pattern Recognition

Traditional intrusion detection models often fail to capture the complex interdependencies between network attacks, as they treat threats as independent instances [45]. However, in IoT environments, attacks often exhibit structured relationships, such as botnet infections or coordinated multi-stage intrusions. GNN provide a powerful solution by modeling these relationships in a graph structure, where nodes represent network traffic instances and edges capture behavioral similarities or dependencies.

In the implementation, graphs are constructed independently within each client and within each data partition after the train–validation–test split. Each node represents one network-flow instance, and each local mini-batch contains up to 128 nodes. For every node, edges are formed with its $k = 5$ nearest neighbors according to cosine similarity in the normalized feature space. Self-loops are included, and the adjacency matrix is normalized as $\hat{A} = D^{-1/2}(A + I)D^{-1/2}$. No edges are created across different clients or between training, validation, and test partitions, thereby preventing test samples from influencing graph-based training.

In FLZT-IDS, an attack graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed, where $\mathcal{V}$ represents network events and $\mathcal{E}$ encodes relationships between them. The adjacency matrix $\mathbf{A} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ defines connections between nodes based on shared features or temporal correlations. The implemented graph encoder uses a single normalized GCN layer with ReLU activation. The resulting graph representation is subsequently concatenated with the temporal representation produced by the Attention-based BiLSTM, as illustrated in Figure 2.

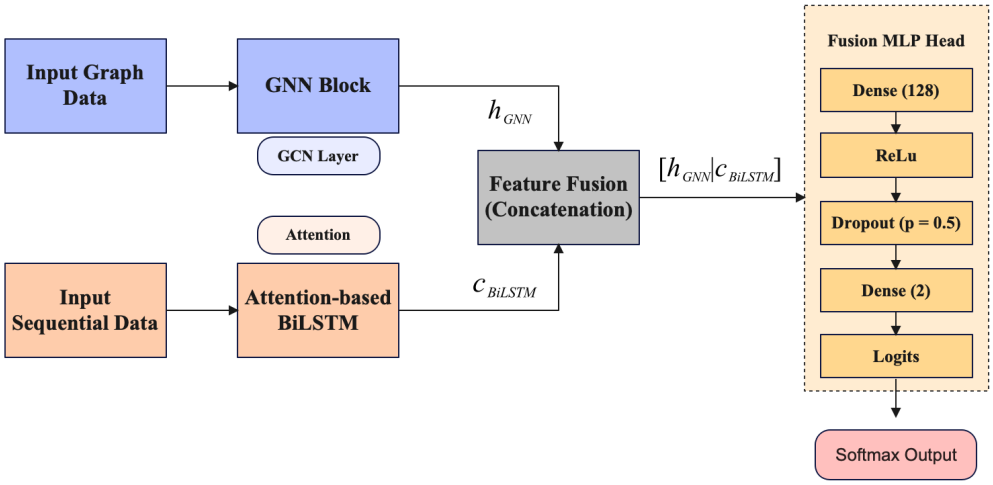


Figure 2. Deep learning architecture in FLZT-IDS. The GCN branch processes graph-structured network-flow data and produces the graph representation h_{GNN}, while the Attention-based BiLSTM processes temporal sequences and produces the attention-weighted context vector c_{BiLSTM}. The two representations are concatenated and passed through a 128-unit dense layer with ReLU activation, dropout ($p = 0.5$), and a two-unit softmax output layer.

To enhance the representation of network anomalies, a graph attention mechanism assigns importance scores α_{ij} to neighboring nodes, refining information propagation:

$$\alpha_{ij} = \frac{\exp\left(\mathrm{LeakyReLU}\left(a^{\top}[h_i \| h_j]\right)\right)}{\sum_{k \in \mathcal{N}(i)} \exp\left(\mathrm{LeakyReLU}\left(a^{\top}[h_i \| h_k]\right)\right)} \tag{6}$$

where h_i and h_j are feature embeddings of nodes i and j. Symbol a is a trainable attention vector, and $\mathcal{N}(i)$ represents the neighborhood of node i.

The graph encoder produces a graph-level representation h_{GNN}, which is not classified independently. Instead, h_{GNN} is passed to the feature-fusion stage and combined with the temporal representation produced by the Attention-based BiLSTM.

By incorporating GNN-based relational learning, FLZT-IDS effectively detects attack patterns that traditional IDS models overlook, improving detection accuracy and resilience to sophisticated cyber threats.

3.4. Attention-Based BiLSTM for Sequential Anomaly Detection

Intrusion detection in IoT networks requires analyzing network traffic patterns over time to detect sequential anomalies, such as slow-propagating malware and coordinated attacks. Traditional IDS models struggle to capture long-range dependencies in time-series network data. Bidirectional Long Short-Term Memory (BiLSTM) addresses this limitation by processing each sequence in both forward and backward directions, thereby capturing contextual dependencies from preceding and succeeding network events [46]. To further improve feature representation, an attention mechanism is integrated to emphasize the time-steps that contribute most strongly to anomaly detection.

Given a sequence of network events $\mathbf{X} = \{x_1, x_2, \ldots, x_T\}$, the BiLSTM consists of forward and backward LSTM units. For each direction $d \in \{\rightarrow, \leftarrow\}$, the input gate, forget gate, output gate, candidate memory state, cell state, and hidden state are computed as:

$$i_t^d = \sigma\left(W_i^d x_t + U_i^d h_{t-1}^d + b_i^d\right), \tag{7}$$

$$f_t^d = \sigma\left(W_f^d x_t + U_f^d h_{t-1}^d + b_f^d\right), \tag{8}$$

$$o_t^d = \sigma\left(W_o^d x_t + U_o^d h_{t-1}^d + b_o^d\right), \tag{9}$$

$$\tilde{c}_t^d = \tanh\left(W_c^d x_t + U_c^d h_{t-1}^d + b_c^d\right), \tag{10}$$

$$c_t^d = f_t^d \odot c_{t-1}^d + i_t^d \odot \tilde{c}_t^d, \tag{11}$$

$$h_t^d = o_t^d \odot \tanh\left(c_t^d\right), \tag{12}$$

where i_t^d, f_t^d, and o_t^d denote the input, forget, and output gates, respectively; $\tilde{c}_t^d$ is the candidate memory state; c_t^d is the cell state; and h_t^d is the hidden state in direction d. The matrices $W_{\{\cdot\}}^d$ and $U_{\{\cdot\}}^d$ are learnable input and recurrent weight matrices, respectively, while $b_{\{\cdot\}}^d$ represents the corresponding bias terms. The sigmoid function $\sigma(\cdot)$ controls the gating operations, $\tanh(\cdot)$ produces the candidate and output activations, and $\odot$ denotes element-wise multiplication. The backward LSTM applies the same operations while processing the sequence in reverse temporal order.

The bidirectional representation at time-step t is obtained by concatenating the forward and backward hidden states:

$$h_t = \left[\overrightarrow{h_t} \,\|\, \overleftarrow{h_t}\right]. \tag{13}$$

For the experiments, network flows are ordered by their timestamp within each collection session and converted into sequences of $T = 20$ consecutive flows. Sequences are generated using a sliding window with a stride of 10 flows, corresponding to 50% overlap. Windows do not cross session, client, or train–validation–test boundaries; therefore, temporal sequences used for training contain only local training observations.

To enhance the model's focus on important time-steps, an attention mechanism is applied, computing attention scores α_t for each hidden state:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^{T} \exp(e_j)} \tag{14}$$

where

$$e_t = v^\top \tanh(W_h h_t + b_h) \tag{15}$$

where W_h and b_h are trainable attention parameters and v is the attention vector. The final context vector c, representing the weighted sum of hidden states, is

$$c = \sum_{t=1}^{T} \alpha_t h_t \tag{16}$$

The attention-weighted context vector c_{BiLSTM} is not classified independently. Instead, it is concatenated with the graph representation h_{GNN} to form the fused representation:

$$h_{\text{fusion}} = h_{\text{GNN}} \| c_{\text{BiLSTM}}, \tag{17}$$

where $\|$ denotes vector concatenation. The fused representation is then passed through a 128-unit fully connected layer with ReLU activation and dropout with $p = 0.5$:

$$h_f = \text{Dropout}\Big(\text{ReLU}\Big(W_f h_{\text{fusion}} + b_f\Big), p = 0.5\Big). \tag{18}$$

The final two-class output logits are computed as

$$o = W_o h_f + b_o, \tag{19}$$

and the probability of output class k is obtained using softmax:

$$P(y = k \mid h_f) = \frac{\exp(o_k)}{\sum_{j=1}^{2} \exp(o_j)}, \qquad k \in \{1, 2\}. \tag{20}$$

The two output classes correspond to *Benign* and *Attack*.

By combining the graph-based representation with the attention-weighted BiLSTM representation, FLZT-IDS jointly captures relational and temporal patterns in network traffic. The fused representation is subsequently used for binary intrusion classification.

The overall DL architecture combining graph-based and sequence-based modeling in FLZT-IDS is illustrated in Figure 2.

3.5. Secure Aggregation with Local Differential Privacy (SALDP)

While FL enhances privacy by keeping data local, the transmission of model updates still poses risks of inference attacks and adversarial manipulations. To counter this, FLZT-IDS employs SALDP, ensuring that model updates remain confidential while preventing malicious contributions.

3.5.1. Local Differential Privacy for Secure Model Updates

To provide a bounded sensitivity for the Gaussian mechanism, each client update Δw_i is first clipped using an ℓ_2 norm bound $C = 1.0$:

$$\overline{\Delta w}_i = \frac{\Delta w_i}{\max\Big(1, \frac{\|\Delta w_i\|_2}{C}\Big)}. \tag{21}$$

Gaussian noise is then added to the clipped update:

$$\widetilde{\Delta w}_i = \overline{\Delta w}_i + \mathcal{N}\Big(0, \sigma^2 C^2 I\Big), \tag{22}$$

where σ denotes the Gaussian noise multiplier. With the clipping norm fixed at $C = 1.0$, the ℓ_2 sensitivity of each bounded client update is taken as $\Delta_2 = C = 1.0$. Unlike pure ε-differential privacy, the Gaussian mechanism is characterized using (ε, δ)-differential privacy. For two neighboring updates u and u', the randomized mechanism $\mathcal{M}$ satisfies

$$\Pr[\mathcal{M}(u) \in S] \leq e^{\varepsilon} \Pr[\mathcal{M}(u') \in S] + \delta, \tag{23}$$

for any measurable output set S.

In the federated setting, two clients are randomly sampled from the ten available clients in each communication round, corresponding to a client sampling rate of $q = 0.2$. The privacy failure probability is fixed at $\delta = 10^{-5}$. Privacy loss is accumulated over the complete training process using a Rényi Differential Privacy (RDP) accountant, which accounts for the sampling rate, Gaussian noise multiplier, and composition across 100 com-

munication rounds. The resulting (ε, δ) privacy budgets for the evaluated noise levels are reported in Section 5.3.

3.5.2. Secure Weighted Aggregation for Privacy-Preserving FL

To prevent the aggregation server from accessing individual client updates, FLZT-IDS employs secure weighted aggregation based on additive homomorphic encryption. A common public key pk is made available to participating clients, while the corresponding private key sk is retained by a separate trusted key holder and is not accessible to the aggregation server. After applying local differential privacy, each eligible client i encrypts its weighted perturbed model update as

$$c_i = \mathrm{Enc}_{pk}(|D_i|\widetilde{w}_i), \tag{24}$$

where $|D_i|$ denotes the number of local training samples and $\widetilde{w}_i$ is the locally perturbed model update. Let $\mathcal{S}_t$ denote the set of trusted and active clients whose updates satisfy the communication threshold and are successfully received during communication round t. Using the additive homomorphic property, the server computes

$$C_t = \bigoplus_{i \in \mathcal{S}_t} c_i = \mathrm{Enc}_{pk}\left(\sum_{i \in \mathcal{S}_t} |D_i|\widetilde{w}_i \right), \tag{25}$$

without decrypting any individual client update. Only the aggregated ciphertext C_t is submitted to the trusted key holder for decryption. The resulting aggregate is then normalized to obtain the global model:

$$w^{t+1} = \frac{\mathrm{Dec}_{sk}(C_t)}{\sum_{i \in \mathcal{S}_t} |D_i|}. \tag{26}$$

This procedure is equivalent to weighted FedAvg over the participating clients while preserving the confidentiality of individual updates. Client dropout is handled by constructing $\mathcal{S}_t$ only from trusted clients whose encrypted updates are successfully received within the current communication round; therefore, both the encrypted numerator and the normalization denominator are computed over the same active-client set.

The threat model assumes that the aggregation server and the trusted key holder do not collude. The aggregation server possesses only the public key and is therefore permitted to perform homomorphic operations but not to decrypt individual ciphertexts. The key holder decrypts only the aggregated ciphertext and does not receive individual client ciphertexts. Consequently, FLZT-IDS does not claim protection against collusion between the aggregation server and the key holder; supporting such a stronger threat model would require threshold decryption or a secure multi-party computation protocol.

3.5.3. Robustness Through Trust-Based Client Filtering

FLZT-IDS addresses adversarial client participation before secure aggregation rather than applying a second aggregation rule after decryption. Each participating client is evaluated using the dynamic trust score defined in Equation (2), and clients with $\tau_i \leq \tau_{\text{threshold}}$ are excluded from the active aggregation set $\mathcal{S}_t$. Gradient-deviation monitoring further identifies abnormal local training behavior before an update is accepted for transmission. Consequently, only updates from clients that satisfy the trust, gradient-deviation, and communication criteria are included in the secure weighted aggregation procedure described in Section 3.5.2.

This design separates adversarial filtering from aggregation: trust-based verification determines which client updates are eligible, while a single secure weighted FedAvg

operation combines the accepted updates. No coordinate-wise median or individual-update decryption is performed at the server.

3.6. Model Training & Communication Rounds

The training process in FLZT-IDS follows a federated and adaptive communication strategy to balance model convergence and communication efficiency. Each IoT device i trains a local model w_i using its dataset D_i, updating parameters via gradient descent:

$$w_i^{t+1} = w_i^t - \eta \nabla L_i(w_i^t) \tag{27}$$

where η is the learning rate.

To reduce communication overhead, updates are transmitted only when the model change exceeds a threshold ϵ_{comm}:

$$||w_i^{t+1} - w_i^t|| > \epsilon_{\text{comm}} \tag{28}$$

The accepted client updates are combined using the secure weighted FedAvg procedure described in Section 3.5.2. Trust-based participation and gradient-deviation monitoring are applied before aggregation, while the server operates only on encrypted accepted updates and does not access individual plaintext model updates. This provides a single and consistent aggregation procedure across the FLZT-IDS framework. The complete workflow of the proposed FLZT-IDS framework is summarized in Algorithm 1, and its visual execution process is illustrated in Figure 3.

Algorithm 1 FLZT-IDS: Federated Learning with Zero-Trust and Secure Aggregation

Input: Global model w^0, rounds T, devices $\{1, \ldots, N\}$, thresholds $\tau_{\text{threshold}}$, $\delta_{\max}$, ϵ_{comm}, privacy noise scale σ, learning rate η, public key pk
Output: Trained global model w^T
for $t = 0$ to $T - 1$ **do**
  Randomly sample client set $\mathcal{C}_t$ with $|\mathcal{C}_t| = 2$ $(q = 0.2)$
  Initialize active-client set $\mathcal{S}_t \leftarrow \emptyset$
  for all device $i \in \mathcal{C}_t$ **in parallel do**
    Compute trust score $\tau_i = \alpha R_i + \beta C_i + \gamma A_i$
    if $\tau_i > \tau_{\text{threshold}}$ **then**
      Initialize $w_i^t \leftarrow w^t$ and train local model
      Compute gradient deviation δ_i
      if $\delta_i > \delta_{\max}$ **then**
        Continue local training
      end if
      if $\|w_i^{t+1} - w_i^t\| > \epsilon_{\text{comm}}$ **then**
        Add noise: $\widetilde{w}_i = w_i^{t+1} + \mathcal{N}(0, \sigma^2)$
        Encrypt weighted update: $c_i = \text{Enc}_{pk}(|D_i|\widetilde{w}_i)$
        Add i to $\mathcal{S}_t$
      end if
    end if
  end for
  if $\mathcal{S}_t \neq \emptyset$ **then**
    Homomorphically aggregate: $C_t = \bigoplus_{i \in \mathcal{S}_t} c_i$
    Decrypt aggregate: $z_t = \text{Dec}_{sk}(C_t)$
    Update global model: $w^{t+1} = z_t / \sum_{i \in \mathcal{S}_t} |D_i|$
  else
    $w^{t+1} \leftarrow w^t$
  end if
end for
return w^T

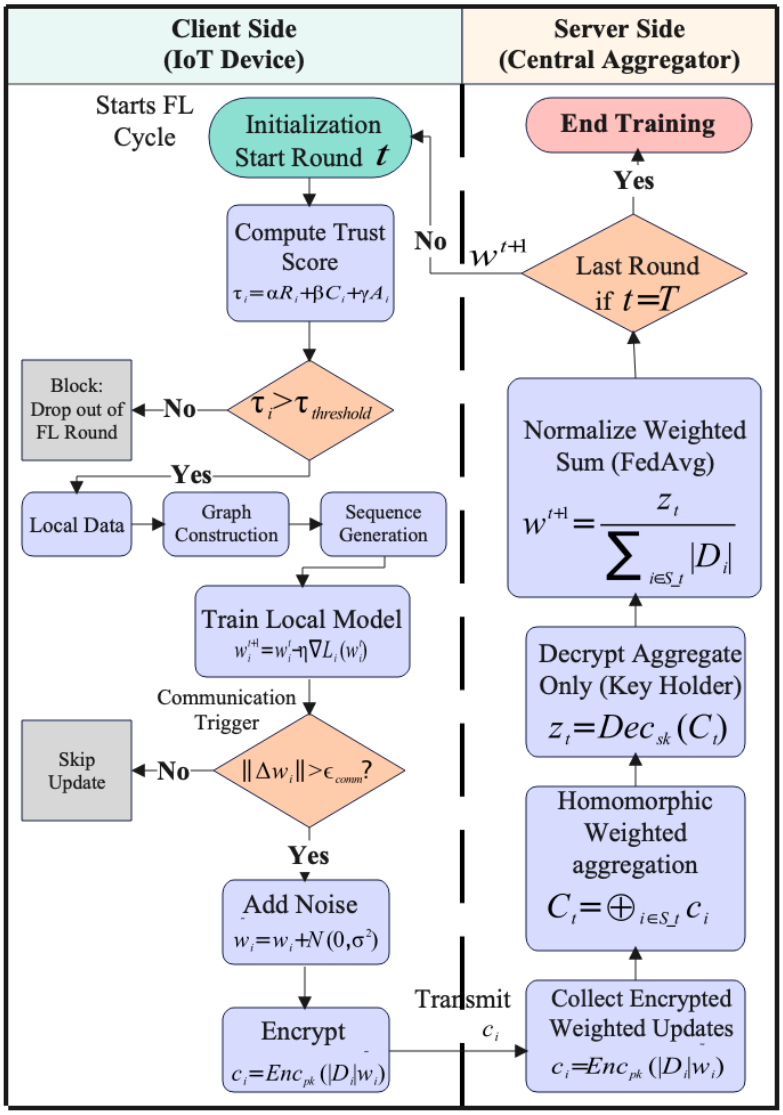


Figure 3. Execution flow of FLZT-IDS showing the client–server boundary, local graph and sequence construction, gradient-deviation screening, communication triggering, privacy perturbation, encrypted model-update transmission, secure aggregation, aggregate-only decryption, and global-model update. Plaintext local model updates remain on the client side, while only encrypted perturbed updates are transmitted to the aggregation server.

4. Experimental Setup

This section outlines the experimental design used to assess the effectiveness of the proposed FLZT-IDS framework. It includes details on the datasets, implementation environment, performance evaluation metrics, and the baseline models used for comparison.

4.1. Datasets Used

To evaluate the intrusion detection capability of FLZT-IDS under both general and industrial IoT scenarios, we employ two widely used and comprehensive benchmark datasets:

(1) CICIDS2017 [47]: Developed by the Canadian Institute for Cybersecurity, this dataset comprises a wide range of real-world attack scenarios such as DDoS, brute-force, botnet, infiltration, and port scanning, collected in a realistic network setting. Each network flow consists of 80+ statistical and behavioral features extracted using CICFlowMeter, making it suitable for training deep learning models for intrusion detection.

(2) Edge-IIoTset [48]: A recent dataset tailored for Industrial IoT (IIoT) environments, Edge-IIoTset includes data from smart grid and factory devices, covering both normal and malicious traffic. It contains various attacks such as ransomware, DDoS, and backdoor exploits, making it highly relevant for evaluating IDS models in edge-computing-based IoT environments.

Although both datasets contain multiple attack categories, this study formulates intrusion detection as a binary classification task by mapping all malicious traffic to the *Attack* class and normal traffic to the *Benign* class. Therefore, the Dense(2) output layer corresponds to these two classes.

Preprocessing and Data Partitioning: Both datasets undergo a unified preprocessing pipeline. Missing values are imputed or removed, categorical attributes are one-hot encoded, and numerical features are normalized using Min–Max scaling. To prevent information leakage, the train, validation, and test partitions are created before oversampling and client allocation. Flows originating from the same collection session are kept within the same partition so that adjacent or closely related network flows do not appear simultaneously in the training and test sets. The resulting data are split into 80% training, 10% validation, and 10% testing subsets using a fixed random seed of 42. Min–Max scaling parameters are fitted exclusively on the training data and then applied unchanged to the validation and test sets. After the Dirichlet allocation, SMOTE and majority-class under-sampling are performed independently within each client's local training data; no resampling is applied to the validation or test sets [49].

The federated non-IID setting is constructed using label-distribution skew. The training data are distributed among 10 simulated clients using a Dirichlet distribution with concentration parameter $\alpha_{\mathrm{Dir}} = 0.5$. Smaller values of α_{Dir} produce more heterogeneous class distributions across clients. The validation and test sets are not Dirichlet-partitioned and remain fixed across all experiments to ensure a consistent evaluation of the global model. The resulting dataset sizes and train–validation–test partitions are summarized in Table 3. The client-level class proportions resulting from the Dirichlet-based non-IID allocation are reported in Table 4, illustrating the heterogeneous label distribution across the 10 federated clients.

Table 3. Dataset partitioning used in the experiments.

Dataset	Total	Train (80%)	Validation (10%)	Test (10%)
CICIDS2017	2,830,743	2,264,594	283,074	283,075
Edge-IIoTset	2,219,201	1,775,361	221,920	221,920

Table 4. Class distribution across the 10 federated clients under Dirichlet label skew ($\alpha_{\mathrm{Dir}} = 0.5$).

Client	Benign (%)	Attack (%)
1	78.4	21.6
2	34.7	65.3
3	62.1	37.9
4	18.9	81.1
5	71.5	28.5
6	43.8	56.2
7	84.2	15.8
8	29.6	70.4
9	55.3	44.7
10	39.4	60.6

Table 3 reports the dataset-level sample counts before local resampling, while Table 4 reports the client-level benign and attack distributions prior to local balancing. Since SMOTE and undersampling are applied independently within each client's training subset after non-IID partitioning, the validation and test sets remain unchanged and are not involved in any resampling operation.

4.2. Implementation Details

FLZT-IDS is implemented in Python 3.11.9 using PyTorch 2.4.1, PyTorch Geometric 2.6.1, and Flower 1.11.1 [50]. Model training uses the Adam optimizer with a learning rate of 0.01, weight decay of 1×10^{-5}, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. Model parameters are initialized using Xavier uniform initialization for the learnable linear and graph-convolutional layers, while recurrent weights in the BiLSTM use orthogonal initialization. Bias terms are initialized to zero. A fixed random seed of 42 is used for Python, NumPy, and PyTorch to improve reproducibility.

Experiments are executed on an Apple MacBook Pro with an M3 chip, 8-core CPU, 10-core GPU, and 16 GB RAM running macOS Sonoma 14.x. PyTorch's Metal Performance Shaders (MPS) backend is used for hardware acceleration during model training, with CPU execution used for operations not supported by MPS. The model is trained for 100 global communication rounds across 10 simulated clients under non-IID settings, with five local epochs per round and a batch size of 128. The complete simulation parameters and system configuration are summarized in Table 5.

Table 5. Simulation parameters and system configuration for FLZT-IDS.

Parameter	Value
Operating System	macOS Sonoma 14.x
Frameworks	PyTorch, PyTorch Geometric, Flower
Learning Rate	0.01
Batch Size	128
Epochs per Round	5
Number of Clients	10
Update Clipping Norm (C)	1.0
Client Sampling Rate (q)	0.2 (2 of 10 clients/round)
Privacy Failure Probability (δ)	10^{-5}
Privacy Accountant	RDP Accountant
Default Privacy Noise Scale (σ)	0.8
Communication Threshold (ϵ_{comm})	0.01
Trust Threshold ($\tau_{\text{threshold}}$)	0.6
Trust Weights (α, β, γ)	(0.4, 0.3, 0.3)
Trust Evidence Window (W)	5 communication rounds
Initial Trust Score (τ_i^0)	0.7
Global Rounds	100

4.3. Performance Metrics

To evaluate the effectiveness of FLZT-IDS, we employ a combination of traditional classification metrics and system-level evaluation criteria. These metrics are designed to capture not only the detection capability of the model but also its robustness against adversarial behaviors, privacy tradeoffs, and communication efficiency in federated IoT environments.

For this binary classification setting, the Attack class is treated as the positive class and the Benign class as the negative class; therefore, Precision, Recall, and F1-score are reported for the positive Attack class rather than as macro-, micro-, or weighted-average multiclass measures.

- **Accuracy**: Measures the overall correctness of the model by computing the ratio of correctly classified samples (both benign and malicious) to the total number of samples. This provides a high-level view of model performance but can be misleading in imbalanced datasets.
- **Precision**: Indicates the proportion of true positive attack detections among all samples predicted as attacks. A high precision score reflects a low false alarm rate and is particularly important for minimizing unnecessary responses to benign activities.
- **Recall**: Measures the proportion of actual attacks correctly identified by the model. A high recall ensures that the system can effectively detect intrusions and minimize the likelihood of undetected threats.
- **F1-Score**: Serves as the harmonic mean of precision and recall, offering a balanced metric that is especially useful in the presence of class imbalance. It provides a single score that reflects both false positives and false negatives.
- **False Positive Rate (FPR)**: Quantifies the proportion of benign samples incorrectly classified as attacks. A low FPR is essential to maintain trust in the IDS and avoid alert fatigue in practical deployments.
- **Communication Overhead**: Represents the average volume (in kilobytes) of model updates transmitted per client in each communication round. This metric evaluates the scalability and communication efficiency of FLZT-IDS, which is critical for resource-constrained IoT networks.

Beyond these traditional evaluation metrics, we conduct targeted experimental analyses to validate specific components of the proposed architecture, including its privacy-preserving and adversarial-resilient capabilities:

- **Privacy–Utility Tradeoff**: We measure the detection accuracy of FLZT-IDS as the LDP noise scale (σ) varies. This quantifies the impact of added noise on model performance and validates the effectiveness of our SALDP scheme.
- **Adversarial Robustness**: To evaluate the resilience of FLZT-IDS against malicious clients, we simulate model poisoning scenarios where a subset of clients submit manipulated updates. The model's accuracy degradation under increasing attack rates is observed and compared with baseline models lacking Zero-Trust validation.
- **Trust Filtering Impact**: We assess how varying the trust threshold $\tau_{\text{threshold}}$ affects the inclusion or exclusion of clients during the FL process. This experiment demonstrates the effectiveness of the Zero-Trust module in filtering unreliable participants and its impact on detection quality and false positives.
- **Communication Efficiency:** We evaluate the communication overhead by measuring the volume of model updates transmitted during federated training. This analysis quantifies the communication savings achieved by the adaptive update and trust-filtering mechanisms.

These metrics collectively ensure a comprehensive and rigorous evaluation of FLZT-IDS across multiple performance dimensions, aligning with the key contributions of this work: privacy preservation, scalability, adversarial robustness, and accuracy.

4.4. Baseline Comparisons

FLZT-IDS is compared against three recent and representative FL-based IDS frameworks that reflect different design priorities:

FL-IDS [40]: A classical FL-based IDS using the FedAvg algorithm and a shallow artificial neural network (ANN). While it supports collaborative learning across distributed nodes, it lacks built-in privacy mechanisms or adversarial filtering strategies.

FedGAT [41]: An attention-based graph neural network model operating in a federated setting. It effectively models relational dependencies in network flows but does not apply differential privacy or incorporate trust-aware participation.

Fed-ANIDS [51]: An anomaly-based IDS that leverages autoencoders in a federated setting. It is designed to preserve user data privacy by avoiding raw data sharing but does not utilize graph structures or Zero-Trust validation.

All models are trained and evaluated under identical experimental settings on CICIDS2017 and Edge-IIoTset datasets. FLZT-IDS consistently demonstrates superior detection capability, efficient communication, and robust defense against adversarial and untrusted participants, as shown in the following results.

5. Results & Analysis

5.1. Ablation Study on Model Components

To assess the contribution of each key component within the FLZT-IDS framework, we perform an ablation study by selectively disabling individual modules, including GNN, Attention-based BiLSTM, SALDP, and the Zero-Trust mechanism. The performance is evaluated on both CICIDS2017 and Edge-IIoTset datasets using standard metrics. The results are summarized in Table 6.

Table 6. Performance of FLZT-IDS under different ablation scenarios on CICIDS2017 and Edge-IIoTset datasets.

Model Variant	CICIDS2017						Edge-IIoTset					
	Accuracy	Precision	Recall	F1	FPR	Comm. (KB)	Accuracy	Precision	Recall	F1	FPR	Comm. (KB)
FLZT-IDS (Full)	97.8%	98.1%	97.5%	97.8%	1.9%	34.2	96.2%	96.9%	95.4%	96.1%	2.3%	33.5
w/o GNN	94.2%	95.3%	92.4%	93.8%	3.5%	33.1	92.6%	93.5%	90.1%	91.8%	4.1%	32.8
w/o BiLSTM	93.6%	94.8%	91.1%	92.9%	3.9%	33.6	91.8%	93.0%	89.2%	91.0%	4.5%	33.2
w/o SALDP	96.5%	97.0%	95.1%	96.0%	2.4%	31.8	94.7%	95.5%	93.2%	94.3%	2.8%	31.4
w/o Zero-Trust	95.1%	95.8%	94.3%	95.0%	3.1%	36.9	93.5%	94.1%	92.2%	93.1%	3.6%	35.7

We observe that the full FLZT-IDS model achieves the highest overall performance, with 97.8% accuracy and 97.8% F1-score on CICIDS2017, and 96.2% accuracy and 96.1% F1-score on Edge-IIoTset. When the GNN component is removed, both datasets show a substantial drop in recall and F1-score, indicating that relational pattern recognition is critical for detecting coordinated and structural attacks.

Eliminating the BiLSTM module results in further reductions in recall, especially for time-dependent attacks. This highlights the importance of modeling sequential dependencies in network traffic.

Disabling SALDP reduces both detection accuracy and precision on the two datasets, while slightly lowering communication overhead due to the removal of privacy-preserving operations. This indicates that, under the evaluated configuration, SALDP provides privacy protection without degrading the overall detection performance of the complete FLZT-IDS framework.

Finally, removing the Zero-Trust component leads to a noticeable rise in FPR and communication overhead, suggesting that trust-aware filtering contributes to in securing the aggregation process and reducing noisy contributions from potentially compromised clients.

The ablation results show that removing individual components changes one or more measured outcomes, indicating that the components contribute differently to detection, privacy, and communication performance.

5.2. Model Accuracy Comparison over Rounds

Figure 4 presents the accuracy progression of FLZT-IDS and three baseline models, FL-IDS, FedGAT, and Fed-ANIDS, evaluated over 100 global communication rounds on both

CICIDS2017 and Edge-IIoTset datasets. Across both datasets, FLZT-IDS demonstrates faster convergence and consistently higher accuracy compared to other models. On CICIDS2017, FLZT-IDS reaches an accuracy of 97.8%, outperforming Fed-ANIDS (94.5%), FedGAT (91.5%), and FL-IDS (89.6%). A similar trend is observed on Edge-IIoTset, where FLZT-IDS achieves 96.2% accuracy, ahead of Fed-ANIDS (89.0%), FedGAT (86.3%), and FL-IDS (85.2%).

The superior performance of FLZT-IDS is attributed to the integration of GNN for structural threat detection, Attention-based BiLSTM for temporal anomaly modeling, and its Zero-Trust framework for filtering untrusted participants. Moreover, the privacy-preserving aggregation techniques do not significantly hinder learning progression, demonstrating the model's efficiency under secure federated settings. These results validate the robustness and adaptability of FLZT-IDS in diverse IoT environments and emphasize its effectiveness in delivering high detection accuracy while preserving privacy and model integrity.

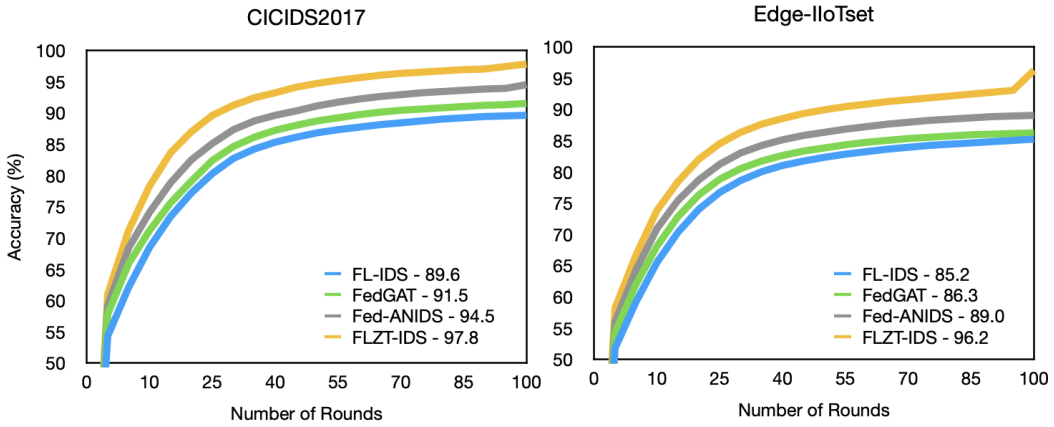


Figure 4. Accuracy progression over communication rounds for FLZT-IDS and baseline models on the CICIDS2017 and Edge-IIoTset datasets.

5.3. Privacy–Utility and Privacy Attack Analysis

Figure 5 evaluates the privacy–utility behavior of FLZT-IDS under different Gaussian noise multipliers. In addition to detection accuracy, the cumulative privacy loss is computed using the RDP accountant with clipping norm $C = 1.0$, client sampling rate $q = 0.2$, privacy failure probability $\delta = 10^{-5}$, and 100 communication rounds. As expected, increasing the noise multiplier provides a stronger privacy guarantee at the expense of some detection utility.

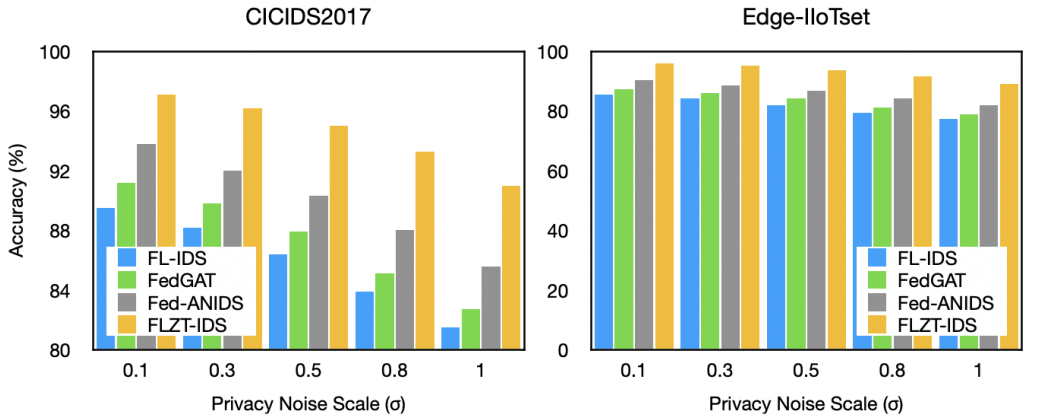


Figure 5. Accuracy comparison of FLZT-IDS and baseline models under varying differential privacy noise scales (σ) on CICIDS2017 and Edge-IIoTset datasets.

Table 7 reports the cumulative privacy budgets obtained after 100 communication rounds. The results demonstrate the expected inverse relationship between the Gaussian noise multiplier and the accumulated privacy loss: larger values of σ produce smaller ε values and therefore stronger differential privacy.

Table 7. Differential privacy accounting after 100 communication rounds ($C = 1.0$, $q = 0.2$, and $\delta = 10^{-5}$).

Noise Multiplier (σ)	δ	Final ε
0.8	10^{-5}	25.6
1.0	10^{-5}	18.2
1.2	10^{-5}	12.9
1.5	10^{-5}	9.3
2.0	10^{-5}	6.2

To complement the analytical privacy accounting, we additionally evaluate resistance to membership inference attacks (MIA). The attacker attempts to determine whether a given sample was included in the training data based on the output confidence of the trained global model. Attack accuracy and the area under the receiver operating characteristic curve (AUC) are reported, where values approaching 50% attack accuracy and an AUC of 0.5 indicate performance close to random guessing. The non-private model is used as the reference setting and is compared with SALDP under progressively stronger Gaussian noise levels.

As shown in Table 8, the non-private model is more susceptible to membership inference, with an attack accuracy of 69.4% and an AUC of 0.741. Applying SALDP progressively reduces the attacker's ability to distinguish training members from non-members. At $\sigma = 0.8$, the attack accuracy decreases to 58.7%, while increasing the noise multiplier to $\sigma = 2.0$ reduces it to 51.6%, with an AUC of 0.519, approaching random guessing. This increased privacy is accompanied by the expected utility trade-off, with model accuracy decreasing from 97.8% without DP to 88.7% at $\sigma = 2.0$. These results complement the analytical (ε, δ) accounting by demonstrating reduced empirical information leakage under progressively stronger privacy settings.

Table 8. Membership inference attack evaluation under different privacy settings on CICIDS2017.

Setting	Attack Accuracy	Attack AUC	Model Accuracy
Without DP	69.4%	0.741	97.8%
SALDP ($\sigma = 0.8$)	58.7%	0.608	94.6%
SALDP ($\sigma = 1.2$)	54.1%	0.555	92.3%
SALDP ($\sigma = 2.0$)	51.6%	0.519	88.7%

5.4. Adversarial Robustness Against Malicious Clients

To evaluate the robustness of FLZT-IDS under adversarial conditions, we simulate an untargeted sign-flipping model-poisoning attack. Malicious clients reverse the direction of their locally computed model updates before transmission, i.e., $\Delta w_i^{adv} = -\Delta w_i$. The attackers have access only to their own local data and model updates and do not know the private data of other clients or the server-side trust scores. We evaluate malicious-client ratios of 0%, 10%, 20%, and 30%, corresponding to zero, one, two, and three malicious clients, respectively, in the 10-client simulation. Figure 6 presents the resulting accuracy trends on CICIDS2017 and Edge-IIoTset.

FLZT-IDS consistently demonstrates superior resilience. On CICIDS2017, while FL-IDS and FedGAT drop to 72.3% and 76.5% respectively at 30% poisoning, FLZT-IDS maintains

an accuracy of 89.3%. Similarly, on Edge-IIoTset, FLZT-IDS retains 87.1% accuracy, significantly higher than Fed-ANIDS (75.0%) and FedGAT (70.3%). This robustness stems from the Zero-Trust-based participant verification mechanism and gradient-deviation screening, which filter potentially corrupted updates before the accepted updates are combined using secure weighted FedAvg. These results demonstrate robustness under the evaluated untargeted poisoning scenario within the 10-client simulation. Because the experiments involve at most three malicious clients, the findings should not be interpreted as a general robustness guarantee against larger-scale, adaptive, or stealthy federated attacks.

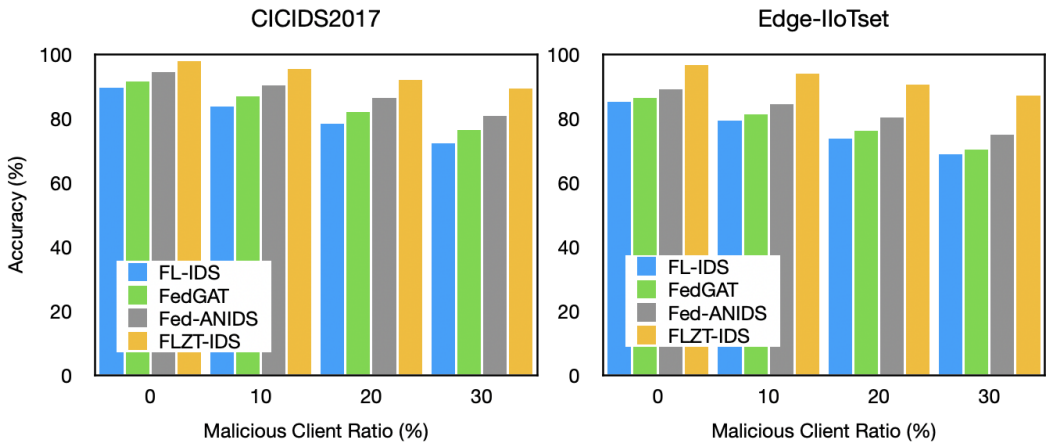


Figure 6. Accuracy degradation of FLZT-IDS and baseline models under varying malicious client ratios.

5.5. Impact of Trust Filtering

To assess the effectiveness of the Zero-Trust mechanism in FLZT-IDS, we examine how varying the trust threshold $\tau_{\text{threshold}}$ affects detection performance. Figure 7 presents accuracy and FPR for different τ values ranging from 0.4 to 0.7 on both CICIDS2017 and Edge-IIoTset datasets.

As shown in subplots (a) and (c), accuracy improves with increasing τ, peaking at $\tau = 0.6$ for both datasets. Beyond this point, accuracy slightly declines, suggesting that excessively strict filtering may exclude benign clients. Specifically, on CICIDS2017 (Figure 7a), accuracy rises from 93.4% at $\tau = 0.4$ to 97.8% at $\tau = 0.6$, then drops slightly to 96.5% at $\tau = 0.7$. On Edge-IIoTset (Figure 7c), a similar trend is observed with peak accuracy at 96.2%.

Correspondingly, subplots (b) and (d) show that FPR decreases as τ increases, demonstrating improved attack detection reliability. For instance, the FPR on CICIDS2017 decreases from 5.8% to 2.7% as τ increases from 0.4 to 0.6, before rising slightly at 0.7 (Figure 7b). The Edge-IIoTset dataset exhibits similar behavior, with the lowest FPR of 3.1% at $\tau = 0.6$ (Figure 7d). At the selected threshold $\tau = 0.6$, the false rejection rate, defined as the proportion of benign participating clients incorrectly excluded by the trust filter, was 4.6% on CICIDS2017 and 5.1% on Edge-IIoTset, indicating that the selected threshold provides a reasonable balance between malicious-client filtering and benign-client participation. These results confirm that the Zero-Trust-based filtering mechanism in FLZT-IDS effectively improves detection accuracy and reduces false alarms, provided the trust threshold is optimally tuned.

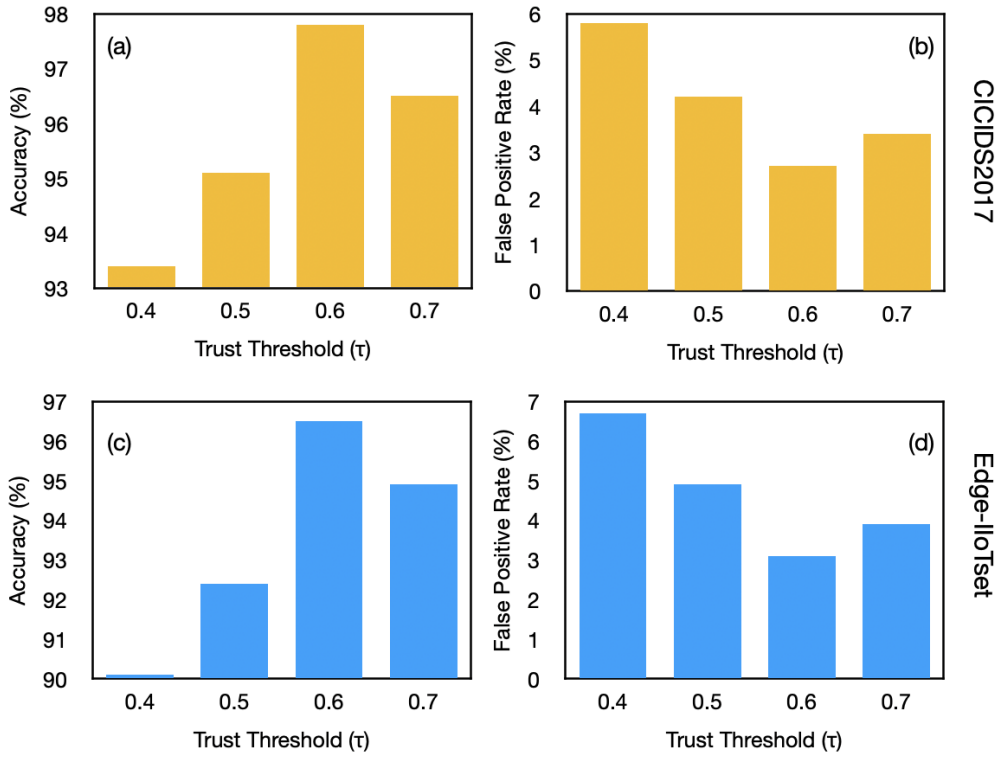


Figure 7. (**a**–**d**) Effect of varying the trust threshold $\tau_{\text{threshold}}$ on accuracy and false positive rate (FPR) in FLZT-IDS.

5.6. Communication Efficiency

Figure 8 compares the cumulative communication overhead incurred by FLZT-IDS against three baseline models, across varying numbers of global rounds (25, 50, 75, and 100). The results on both CICIDS2017 and Edge-IIoTset datasets demonstrate that FLZT-IDS achieves significant reductions in communication cost over time.

This improvement is attributed to two key factors: (i) the integration of an adaptive update mechanism, which limits model transmissions to only significant updates based on the threshold ϵ_{comm}, and (ii) the trust filtering process that excludes low-trust clients, further minimizing unnecessary communication. Compared to FL-IDS, FLZT-IDS reduces cumulative communication overhead by more than 30% after 100 rounds on both datasets, while still achieving superior detection accuracy.

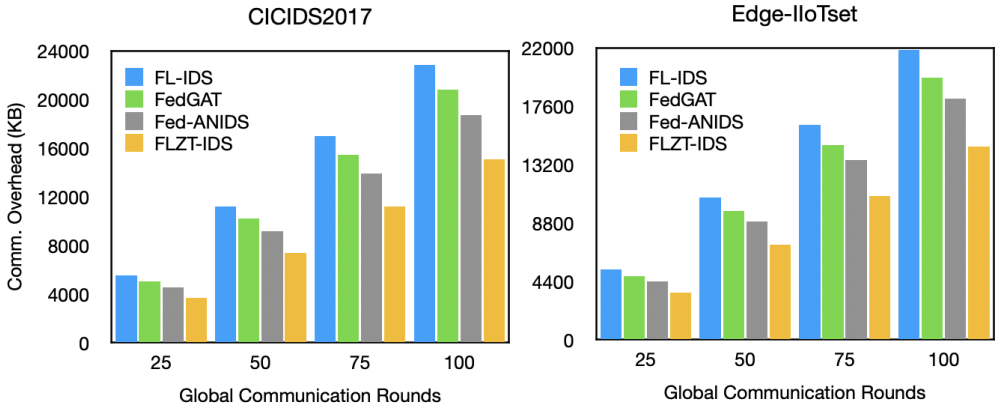


Figure 8. Cumulative communication overhead (in KB) of FLZT-IDS and baseline models across different numbers of global communication rounds on CICIDS2017 and Edge-IIoTset datasets.

6. Discussion

6.1. Strengths & Improvements over Existing Work

The proposed FLZT-IDS framework introduces a comprehensive enhancement over existing federated IDS by simultaneously addressing privacy, trust, efficiency, and adaptability challenges in IoT environments. Unlike conventional FL-based IDS models that rely solely on federated updates, FLZT-IDS integrates a Zero-Trust mechanism to filter out potentially malicious or unreliable clients based on dynamic trust scores. This results in higher model robustness under adversarial conditions, as demonstrated in our poisoning resilience experiments.

From a privacy standpoint, FLZT-IDS incorporates SALDP with bounded update clipping and Gaussian perturbation. Privacy loss is quantified using (ε, δ)-DP and RDP accounting over the complete federated training process, while membership inference experiments provide an empirical assessment of information leakage. The privacy–utility analysis demonstrates the trade-off between stronger privacy protection and intrusion-detection performance.

In terms of detection capability, the hybrid deep learning architecture combining GNN and Attention-based BiLSTM allows the model to capture both structural and sequential attack patterns more effectively. This dual-view learning strategy contributes to consistently higher accuracy and lower FPR on both CICIDS2017 and Edge-IIoTset datasets.

Furthermore, FLZT-IDS demonstrates substantial reductions in communication overhead due to its adaptive update strategy. By limiting unnecessary transmissions, the framework improves communication efficiency in bandwidth-constrained federated IoT environments. However, the present study does not provide a complete computational-cost evaluation in terms of FLOPs, memory usage, inference latency, energy consumption, or cryptographic processing time; these aspects require evaluation on representative resource-constrained edge devices in future work.

6.2. Limitations & Future Directions

Despite its strengths, FLZT-IDS has several limitations that open avenues for future research. First, the current implementation assumes a simulated federated environment with a fixed number of IoT clients. While this setup supports controlled and reproducible evaluation, it may not fully capture the variability of real-world federated networks, where client participation, connectivity, computational capacity, and data distributions can change dynamically.

Second, although the trust-score mechanism filters potentially malicious clients, it relies on predefined indicators such as recent activity, protocol compliance, and anomaly confidence. More sophisticated trust inference mechanisms, including reinforcement learning or reputation-based approaches, could further improve resilience against stealthy or evolving adversaries. Third, the current adversarial evaluation is limited to the specified poisoning scenario and a 10-client simulation. Therefore, the reported robustness results should not be interpreted as a general guarantee against broader classes of attacks. Future work should evaluate adaptive and stealthy attacks and include comparisons with dedicated robust aggregation methods such as Trimmed Mean, Krum, RFA, FLTrust, and FoolsGold.

Fourth, although the proposed model combines GNN and BiLSTM components to improve relational and temporal intrusion detection, the associated computational requirements may remain non-trivial for resource-constrained edge devices. Lightweight model variants, model compression, or hardware-aware neural architecture search could therefore be explored to improve deployment feasibility.

In addition, the current implementation operationalizes Zero-Trust principles through continuous server-side score-based client verification and filtering. It does not constitute a complete identity-management or cryptographic Zero-Trust authentication architecture. Future extensions could incorporate stronger identity verification and authentication mechanisms to provide a more comprehensive Zero-Trust implementation.

Finally, extending FLZT-IDS to support asynchronous federated learning and personalized client models could further improve adaptability in heterogeneous environments, particularly where client availability, data distributions, and device capabilities vary significantly.

7. Conclusions

This paper introduced FLZT-IDS, a Zero-Trust-enabled federated learning framework for secure and privacy-preserving intrusion detection in IoT environments. By integrating GNN and Attention-based BiLSTM, the framework captures both relational and sequential attack patterns. The incorporation of SALDP provides a quantifiable privacy–utility trade-off, supported by cumulative (ε, δ) privacy accounting and membership inference evaluation. Furthermore, the Zero-Trust mechanism filters unreliable clients and improves resilience under the evaluated model-poisoning scenario.

Experiments on CICIDS2017 and Edge-IIoTset show that FLZT-IDS achieves 97.8% and 96.2% accuracy, respectively, while reducing transmitted model-update volume relative to FL-IDS through adaptive communication and trust-based client filtering. Under the evaluated 10-client simulation, the framework also maintains higher detection performance than the selected baselines under the tested poisoning conditions. These findings demonstrate the potential of FLZT-IDS within the controlled experimental setting while recognizing that larger-scale, real-world federated deployments may introduce additional variability in client participation, communication, computational capacity, and adversarial behavior. Future work will therefore focus on broader client populations, real-world edge-device evaluation, stronger adversarial settings, and a more complete assessment of cryptographic and computational overhead.

Author Contributions: Conceptualization, S.K.B.; Methodology, S.K.B. and M.A.; Validation, C.W. and M.A.; Investigation, M.A.; Writing—original draft, S.K.B.; Writing—review & editing, M.A. and C.W.; Supervision, C.W. and S.K.B.; Project administration, C.W. and S.K.B.; Funding acquisition, C.W. and S.K.B., All authors have read and agreed to the published version of the manuscript.

Funding: This work is supported by National Natural Science Foundation of China Project, Project No.: 52577124, and 52207125, and the Key Research and Development Program of Hunan Province Project, Project No.: 2024JK2040 and also Shaoyang University Doctoral Research Start-up Special Project, Project No.: 25KYQD166.

Data Availability Statement: The raw data supporting the conclusions of this article will be made available by the authors on request.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Abasi, A.K.; Aloqaily, M.; Ouni, B.; Guizani, M.; Debbah, M.; Karray, F. A survey on securing 6G wireless communications based optimization techniques. In *Proceedings of the 2023 International Wireless Communications and Mobile Computing (IWCMC)*; IEEE: New York, NY, USA, 2023; pp. 216–223.
2. Rahmani, A.M.; Bayramov, S.; Kiani Kalejahi, B. Internet of things applications: Opportunities and threats. *Wirel. Pers. Commun.* **2022**, *122*, 451–476. [CrossRef] [PubMed]
3. Al-Haija, Q.A.; Droos, A. A comprehensive survey on deep learning-based intrusion detection systems in Internet of Things (IoT). *Expert Syst.* **2025**, *42*, e13726. [CrossRef]

4. Bikila, D.D.; Čapek, J. Machine Learning-Based Attack Detection for the Internet of Things. *Future Gener. Comput. Syst.* **2025**, *166*, 107630. [CrossRef]
5. Lin, Y.D.; Chan, W.H.; Lai, Y.C.; Yu, C.M.; Wu, Y.S.; Lee, W.B. Enhancing can security with ML-based IDS: Strategies and efficacies against adversarial attacks. *Comput. Secur.* **2025**, *151*, 104322. [CrossRef]
6. Friha, O.; Ferrag, M.A.; Shu, L.; Maglaras, L.; Choo, K.K.R.; Nafaa, M. FELIDS: Federated learning-based intrusion detection system for agricultural Internet of Things. *J. Parallel Distrib. Comput.* **2022**, *165*, 17–31. [CrossRef]
7. Yang, H.; Hu, K.; Wang, H.; Hong, W.; Wang, X.; Wang, H.; Song, Y.; Liu, Z. BCLIP-ADer: A Bayesian prompt contrastive language-image pretraining method for catenary component anomaly detection in electrified railways. *IEEE Trans. Transp. Electrif.* **2026**, *12*, 7522–7534.
8. Yan, J.; Chen, B.; Zhang, F.; Cheng, Y.; Wang, H.; Wang, H.; Wang, M.; Li, T.; Zhang, W. Meta-Learning-Based Graph Convolutional Wavelet Network for Intelligent Dynamic Modeling of High-Speed Rail Subsystems. *IEEE Trans. Veh. Technol.* **2026**, *early access*. [CrossRef]
9. Nahar, N.; Andersson, K.; Schelén, O.; Saguna, S. A Survey on Zero Trust Architecture: Applications and Challenges of 6G Networks. *IEEE Access* **2024**, *12*, 94753–94764. [CrossRef]
10. Agrawal, S.; Sarkar, S.; Aouedi, O.; Yenduri, G.; Piamrat, K.; Alazab, M.; Bhattacharya, S.; Maddikunta, P.K.R.; Gadekallu, T.R. Federated learning for intrusion detection system: Concepts, challenges and future directions. *Comput. Commun.* **2022**, *195*, 346–361. [CrossRef]
11. Fedorchenko, E.; Novikova, E.; Shulepov, A. Comparative review of the intrusion detection systems based on federated learning: Advantages and open challenges. *Algorithms* **2022**, *15*, 247. [CrossRef]
12. Asad, M.; Moustafa, A.; Yu, C. A critical evaluation of privacy and security threats in federated learning. *Sensors* **2020**, *20*, 7182. [CrossRef] [PubMed]
13. Mishra, S.; Kshirsagar, V.; Dwivedula, R.; Hota, C. Attention-based bi-lstm for anomaly detection on time-series data. In *Proceedings of the International Conference on Artificial Neural Networks*; Springer: Berlin/Heidelberg, Germany, 2021; pp. 129–140. [CrossRef] [PubMed]
14. Saputra, I.P.; Utami, E.; Muhammad, A.H. Comparison of anomaly based and signature based methods in detection of scanning vulnerability. In *Proceedings of the 2022 9th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI)*; IEEE: New York, NY, USA, 2022; pp. 221–225.
15. AlYousef, M.Y.; Abdelmajeed, N.T. Dynamically detecting security threats and updating a signature-based intrusion detection system's database. *Procedia Comput. Sci.* **2019**, *159*, 1507–1516. [CrossRef]
16. Mutambik, I. An Efficient Flow-Based Anomaly Detection System for Enhanced Security in IoT Networks. *Sensors* **2024**, *24*, 7408. [CrossRef] [PubMed]
17. Ghazi, D.S.; Hamid, H.S.; Zaiter, M.J.; Behadili, A.S.G. Snort Versus Suricata in Intrusion Detection. *Iraqi J. Inf. Commun. Technol.* **2024**, *7*, 73–88. [CrossRef]
18. Kumar, V.; Sinha, D. A robust intelligent zero-day cyber-attack detection technique. *Complex Intell. Syst.* **2021**, *7*, 2211–2234. [CrossRef] [PubMed]
19. Ullah, I.; Mahmoud, Q.H. Design and development of a deep learning-based model for anomaly detection in IoT networks. *IEEE Access* **2021**, *9*, 103906–103926. [CrossRef]
20. Asad, M.; Otoum, S.; Shaukat, S. Clients eligibility-based lightweight protocol in federated learning: An ids use-case. *IEEE Trans. Netw. Serv. Manag.* **2024**, *21*, 3759–3774. [CrossRef]
21. Chauhan, P.; Atulkar, M. An efficient centralized DDoS attack detection approach for Software Defined Internet of Things. *J. Supercomput.* **2023**, *79*, 10386–10422. [CrossRef]
22. Bhavsar, M.; Bekele, Y.; Roy, K.; Kelly, J.; Limbrick, D. FL-IDS: Federated learning-based intrusion detection system using edge devices for transportation IoT. *IEEE Access* **2024**, *12*, 52215–52226. [CrossRef]
23. Alsaleh, S.; Menai, M.E.B.; Al-Ahmadi, S. Federated Learning–Based Model to Lightweight IDSs for heterogeneous IoT Networks: State-of-the-Art, Challenges and Future Directions. *IEEE Access* **2024**, *12*, 134256–134272. [CrossRef]
24. Asad, M.; Moustafa, A.; Ito, T. Fedopt: Towards communication efficiency and privacy preservation in federated learning. *Appl. Sci.* **2020**, *10*, 2864. [CrossRef]
25. Chinamanagonda, S. Zero Trust Security Models in Cloud Infrastructure-Adoption of zero-trust principles for enhanced security. *Acad. Nexus J.* **2022**, *1*, 1–19.
26. Al Ridhawi, I.; Aloqaily, M. Zero-Trust UAV-enabled and DT-supported 6G Networks. In *Proceedings of the GLOBECOM 2023—2023 IEEE Global Communications Conference*; IEEE: New York, NY, USA, 2023; pp. 6171–6176.
27. Ridhawi, I.A.; Otoum, S.; Aloqaily, M. Decentralized zero-trust framework for digital twin-based 6G. *arXiv* **2023**, arXiv:2302.03107.
28. Ismail, M.; Abd El-Gawad, A.F. Revisiting zero-trust security for internet of things. *Sustain. Mach. Intell. J.* **2023**, *3*, 1–8. [CrossRef]
29. Nowroozi, E.; Haider, I.; Taheri, R.; Conti, M. Federated learning under attack: Exposing vulnerabilities through data poisoning attacks in computer networks. *IEEE Trans. Netw. Serv. Manag.* **2025**, *22*, 822–831. [CrossRef]

30. Blika, A.; Palmos, S.; Doukas, G.; Lamprou, V.; Pelekis, S.; Kontoulis, M.; Ntanos, C.; Askounis, D. Federated Learning For Enhanced Cybersecurity And Trustworthiness In 5G and 6G Networks: A Comprehensive Survey. *IEEE Open J. Commun. Soc.* **2024**, *6*, 3094–3130. [CrossRef]
31. Khacha, A.; Saadouni, R.; Harbi, Y.; Aliouat, Z.; Gherbi, C.; Harous, S. Privacy-Preserving in Agricultural IoT: Intrusion Detection Using Federated Learning and CNN. In *Proceedings of the 2024 International Conference on Advanced Aspects of Software Engineering (ICAASE)*; IEEE: New York, NY, USA, 2024; pp. 1–7.
32. Bonawitz, K.; Eichner, H.; Grieskamp, W.; Huba, D.; Ingerman, A.; Ivanov, V.; Kiddon, C.; Konečnỳ, J.; Mazzocchi, S.; McMahan, B.; et al. Towards federated learning at scale: System design. *Proc. Mach. Learn. Syst.* **2019**, *1*, 374–388.
33. Moustafa, A.; Asad, M.; Shaukat, S.; Norta, A. Ppcsa: Partial participation-based compressed and secure aggregation in federated learning. In Proceedings of the International Conference on Advanced Information Networking and Applications, Toronto, ON, Canada, 12–14 May 2021; pp. 345–357.
34. McMahan, B.; Moore, E.; Ramage, D.; Hampson, S.; Y Arcas, B.A. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the Artificial Intelligence and Statistics*; PMLR: New York, NY, USA, 2017; pp. 1273–1282.
35. Znidi, F.; Morsy, M.; Rathore, H. Future on Wheels: Safeguarding Privacy in Tomorrow's Connected Vehicles-FUTURE-SP. *IEEE Access* **2024**, *12*, 179857–179878. [CrossRef]
36. Sobchuk, V.; Pykhnivskyi, R.; Barabash, O.; Korotin, S.; Omarov, S. Sequential intrusion detection system for zero-trust cyber defense of IOT/IIOT networks. *Adv. Inf. Syst.* **2024**, *8*, 92–99. [CrossRef]
37. Asad, M.; Alhasawi, Y. SIAD: A low-latency AI-driven software architecture for anomaly detection in industrial IoT environments. *Clust. Comput.* **2026**, *29*, 427. [CrossRef]
38. Cai, M.; Zhang, Z.; Wu, G.; Chai, T. XMatchAD: A Cross-Modal Matching Perspective on Reconstruction-based Anomaly Detection. *IEEE Trans. Image Process.* **2026**, *35*, 8443–8455. [CrossRef] [PubMed]
39. Soomro, P.D.; Fu, X.; Banbhrani, S.K.; Ali, A.; Asad, M. Federated learning based deep learning framework for robust object detection in indoor and outdoor environments. *Comput. Electr. Eng.* **2026**, *133*, 110910. [CrossRef]
40. Lazzarini, R.; Tianfield, H.; Charissis, V. Federated learning for IoT intrusion detection. *AI* **2023**, *4*, 509–530. [CrossRef]
41. Wu, J.; Qiu, G.; Wu, C.; Jiang, W.; Jin, J. Federated learning for network attack detection using attention-based graph neural networks. *Sci. Rep.* **2024**, *14*, 19088. [CrossRef] [PubMed]
42. He, Y.; Huang, D.; Chen, L.; Ni, Y.; Ma, X. A survey on zero trust architecture: Challenges and future trends. *Wirel. Commun. Mob. Comput.* **2022**, *2022*, 6476274. [CrossRef]
43. Zhang, K.; Sun, J.; Feng, Q.; Huang, X.; Jiang, C.; Hossain, M.S. Joint optimization of fairness and energy efficiency in zero-trust federated learning for consumer internet of things: A lossy communication perspective. *IEEE Trans. Consum. Electron.* **2024**, *71*, 1156–1166. [CrossRef]
44. Li, Z.; He, Y.; Yu, H.; Kang, J.; Li, X.; Xu, Z.; Niyato, D. Data heterogeneity-robust federated learning via group client selection in industrial IoT. *IEEE Internet Things J.* **2022**, *9*, 17844–17857. [CrossRef]
45. Zhang, H.; Cao, T. A Hybrid Approach to Network Intrusion Detection Based On Graph Neural Networks and Transformer Architectures. In *Proceedings of the 2024 14th International Conference on Information Science and Technology (ICIST)*; IEEE: New York, NY, USA, 2024; pp. 574–582.
46. Sivamohan, S.; Sridhar, S.; Krishnaveni, S. An effective recurrent neural network (RNN) based intrusion detection via bi-directional long short-term memory. In *Proceedings of the 2021 International Conference on Intelligent Technologies (CONIT)*; IEEE: New York, NY, USA, 2021; pp. 1–5.
47. Panigrahi, R.; Borah, S. A detailed analysis of CICIDS2017 dataset for designing Intrusion Detection Systems. *Int. J. Eng. Technol.* **2018**, *7*, 479–482.
48. Ferrag, M.A.; Friha, O.; Hamouda, D.; Maglaras, L.; Janicke, H. Edge-IIoTset: A New Comprehensive Realistic Cyber Security Dataset of IoT and IIoT Applications for Centralized and Federated Learning. *IEEE Access* **2022**, *10*, 40281–40306. [CrossRef]
49. Shelke, M.S.; Deshmukh, P.R.; Shandilya, V.K. A review on imbalanced data handling using undersampling and oversampling technique. *Int. J. Recent Trends Eng. Res.* **2017**, *3*, 444–449. [CrossRef]
50. Beutel, D.J.; Topal, T.; Mathur, A.; Qiu, X.; Fernandez-Marques, J.; Gao, Y.; Sani, L.; Li, K.H.; Parcollet, T.; de Gusmão, P.P.B.; et al. Flower: A friendly federated learning research framework. *arXiv* **2020**, arXiv:2007.14390.
51. Idrissi, M.J.; Alami, H.; El Mahdaouy, A.; El Mekki, A.; Oualil, S.; Yartaoui, Z.; Berrada, I. Fed-anids: Federated learning for anomaly-based network intrusion detection systems. *Expert Syst. Appl.* **2023**, *234*, 121000. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.